
Lecture 2.2

Bayesian Phylogenetics I

The Bayesian framework

Bayesian phylogenetic analysis

- Bayesian phylogenetic analysis was developed in the mid 1990s
- Now one of the most widely used methods
- Bayes's theorem (1763)
- Reverend Thomas Bayes



*Image probably
not of Thomas Bayes*

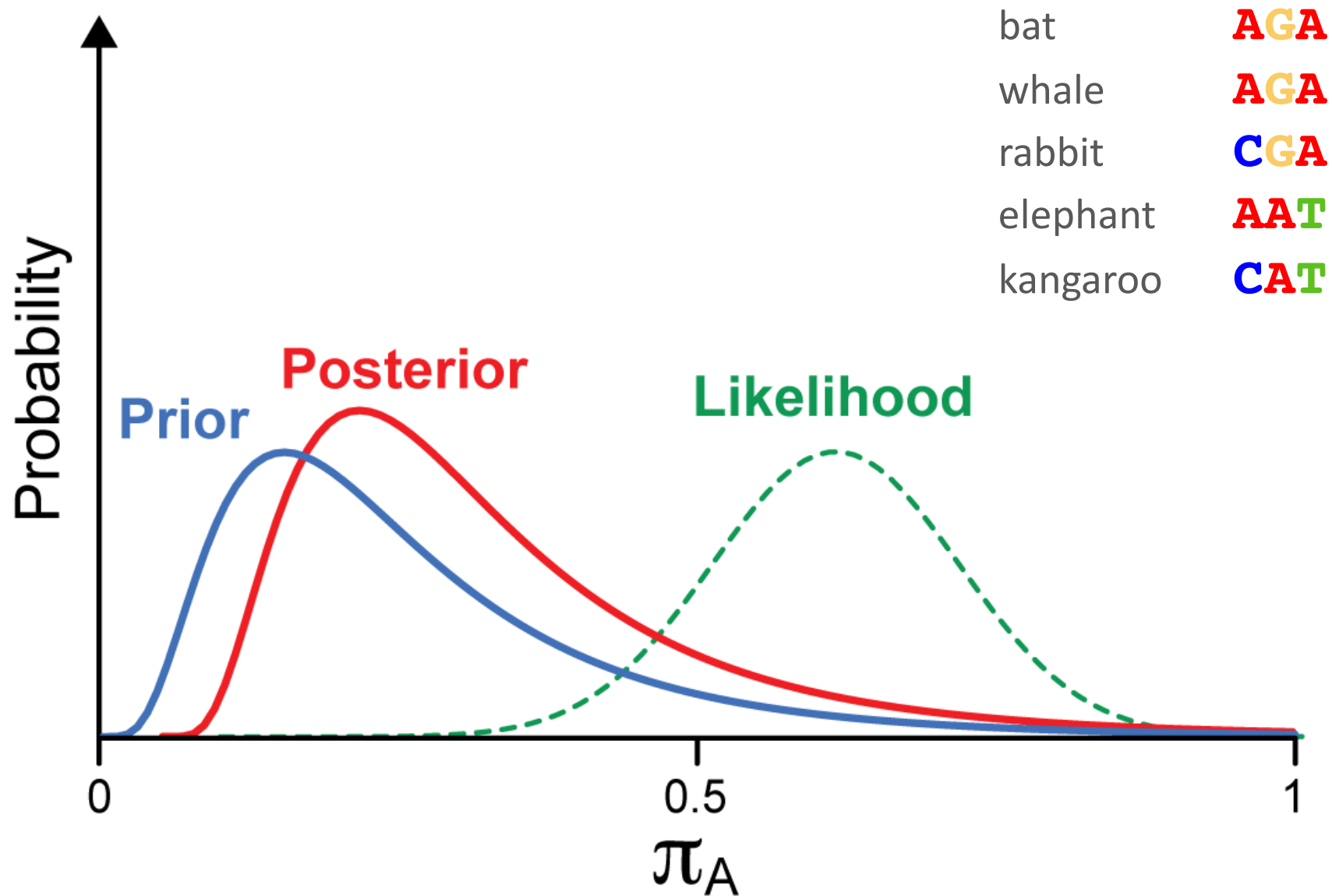
Contrast with frequentist statistics (likelihood)

Bayesian phylogenetic analysis

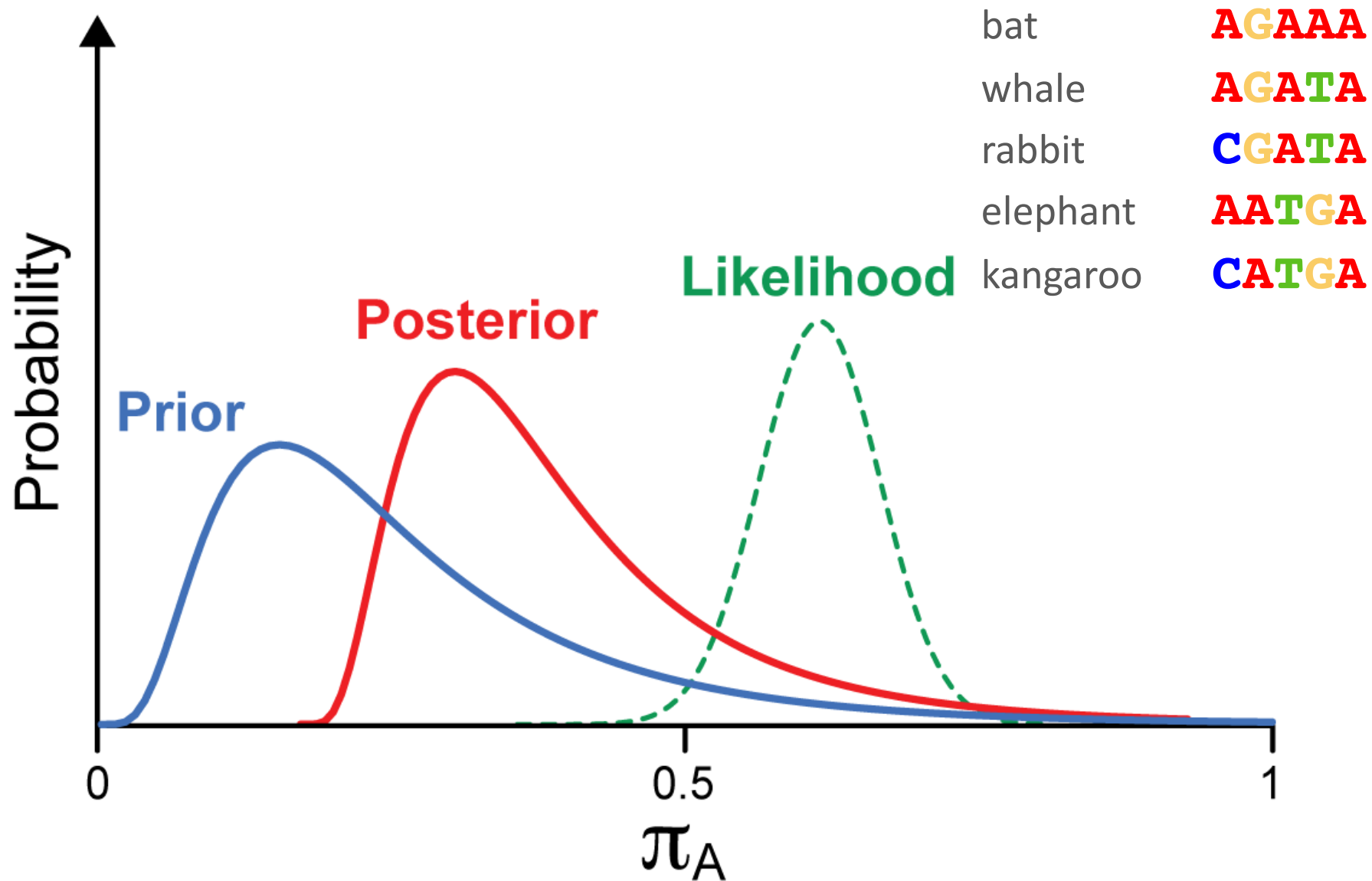
$$\Pr(\theta \mid D) \propto \Pr(\theta) \Pr(D \mid \theta)$$

- Parameters have distributions
- Before the data are observed, each parameter has a prior probability distribution
 - Reflect our prior expectations (and uncertainty) about values of parameters (without knowledge of the data)
- Likelihood of the data is computed
- Prior probability distribution is combined (updated) with the likelihood to yield the posterior probability distribution

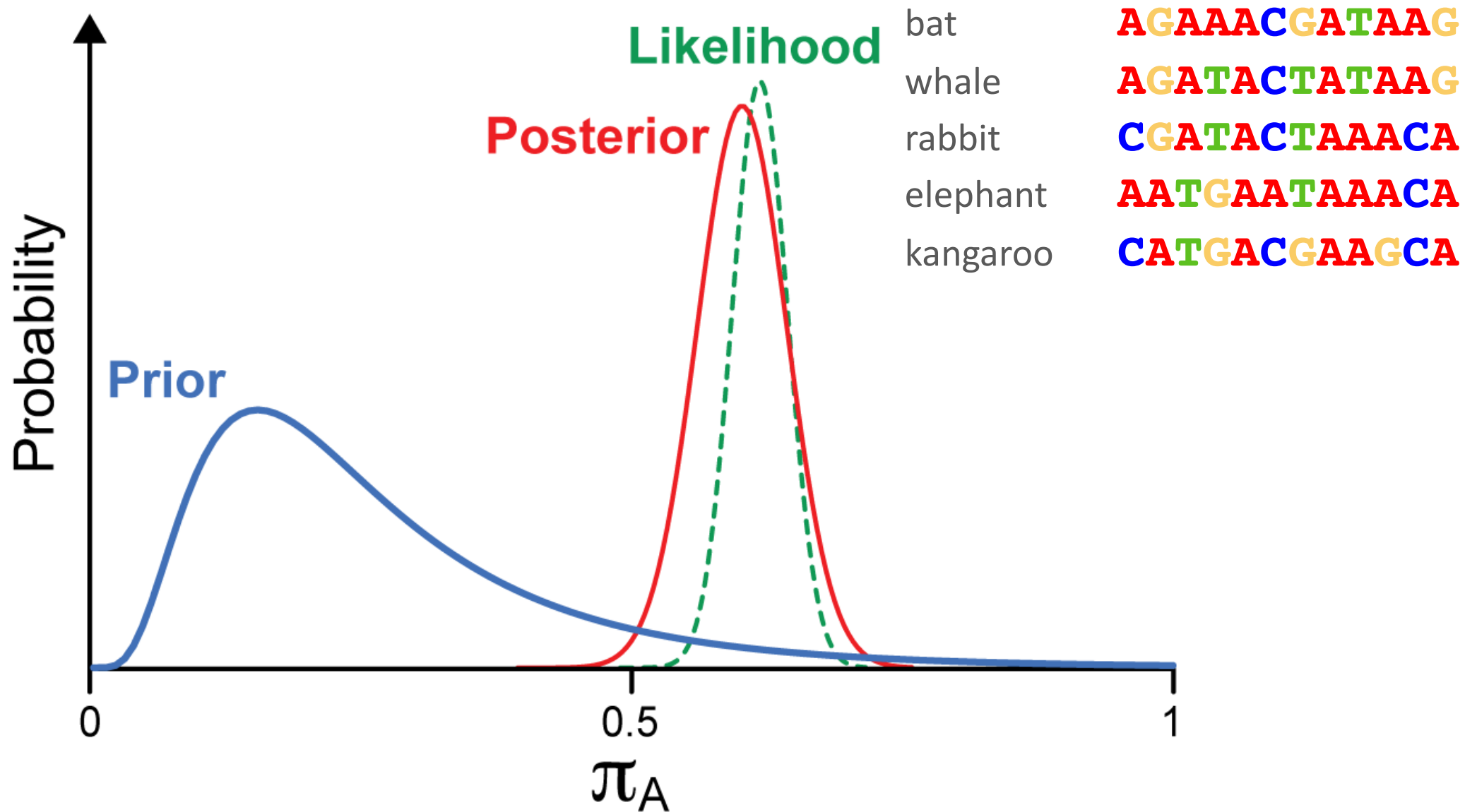
Simple example



Simple example



Simple example



Bayesian inference

Prior

Specified by user,
independent of data

Likelihood

Calculated from data

$$\Pr(\theta | D) = \frac{\Pr(\theta) \Pr(D | \theta)}{\Pr(D)}$$

The diagram shows the equation for Bayesian inference. The numerator consists of two terms, $\Pr(\theta)$ and $\Pr(D | \theta)$, each enclosed in a light blue rounded rectangle. A leader line from the 'Prior' label points to the $\Pr(\theta)$ box, and another leader line from the 'Likelihood' label points to the $\Pr(D | \theta)$ box. The denominator is $\Pr(D)$, also enclosed in a light blue rounded rectangle. A leader line from the 'normalising constant' label points to the $\Pr(D)$ box. The entire equation is centered on the slide.

Posterior

This is what we
want to estimate

normalising constant
marginal likelihood of the data
model likelihood

Bayesian inference

Prior prob of tree

Topology

Branch lengths

**Prior prob of substitution
model parameters**

Rate parameters

Base frequencies

$$\boxed{\Pr(\tau, M \mid D)} = \frac{\boxed{\Pr(\tau)} \boxed{\Pr(M)} \boxed{\Pr(D \mid \tau, M)}}{\Pr(D)}$$

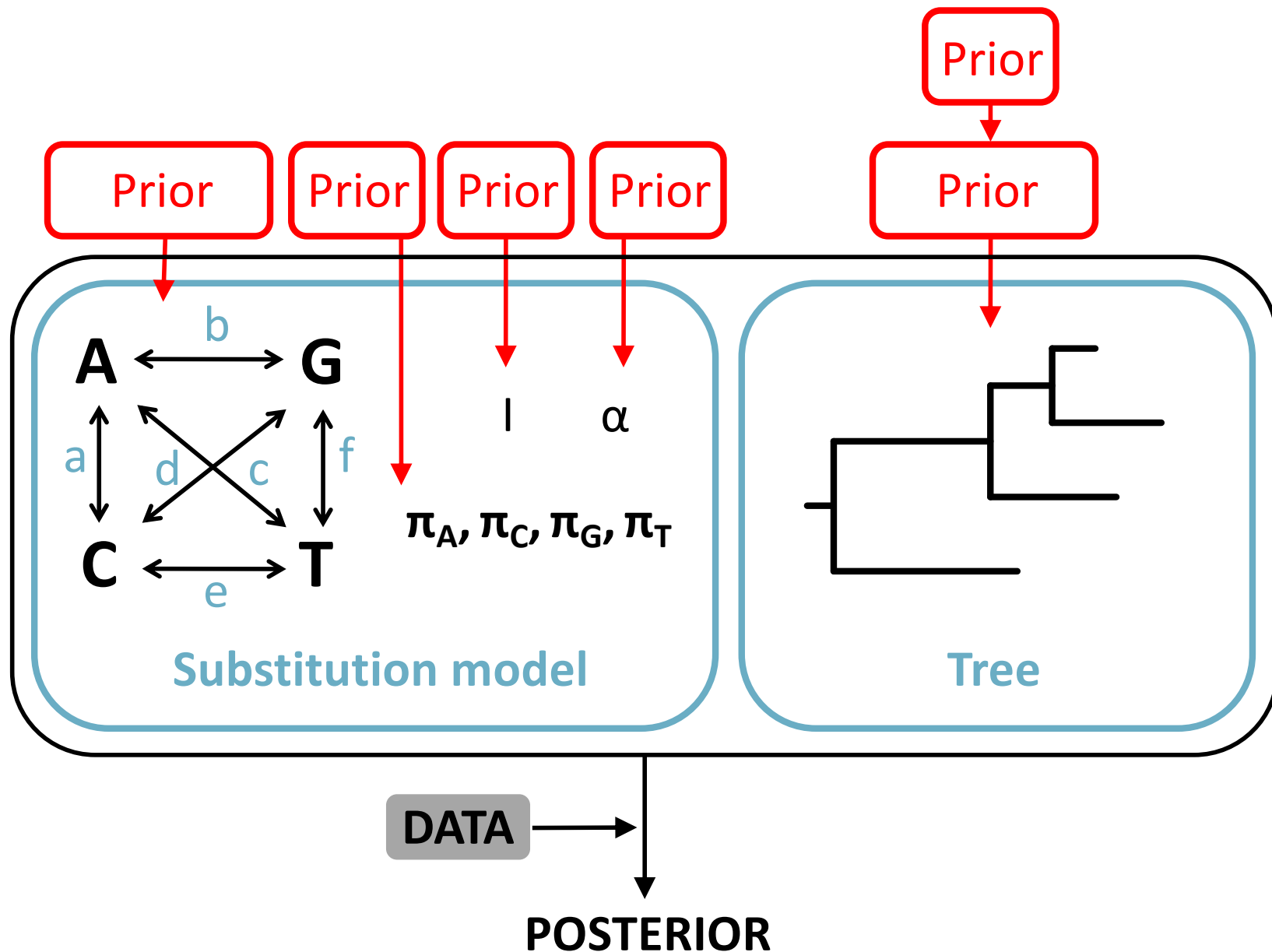
Posterior

This is what we
want to estimate

Likelihood

Calculated from data

Bayesian hierarchical model



Priors

Priors

- Priors are chosen in the form of probability distributions
- Reflect our prior expectations (and uncertainty) about values of parameters (without knowledge of the data)
 - Past observations
 - Personal beliefs
 - Use of a biological model

Continuous distributions

- Uniform
- Normal

Used to specify prior distributions of various continuous parameters

- Exponential
- Lognormal
- Gamma

Used to specify prior distributions of continuous parameters that cannot take negative values

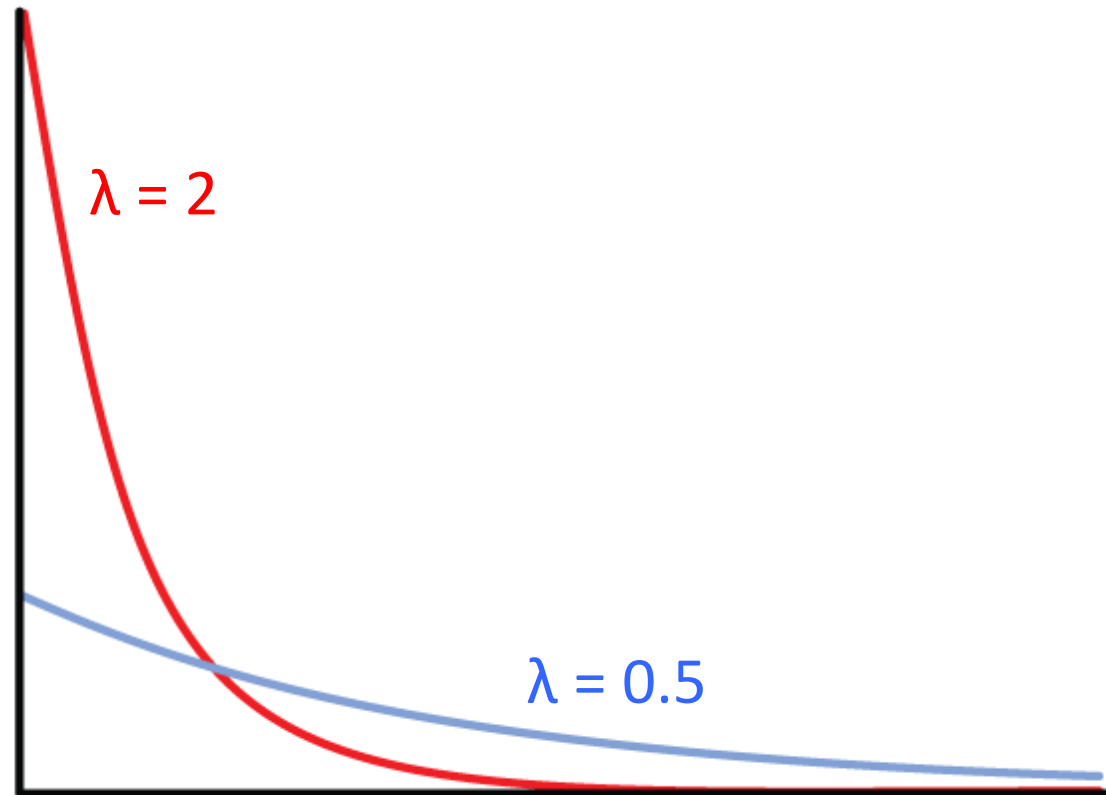
- Beta
- Dirichlet

Continuous distributions

- Uniform
- Normal
- **Exponential**
- Lognormal
- Gamma
- Beta
- Dirichlet

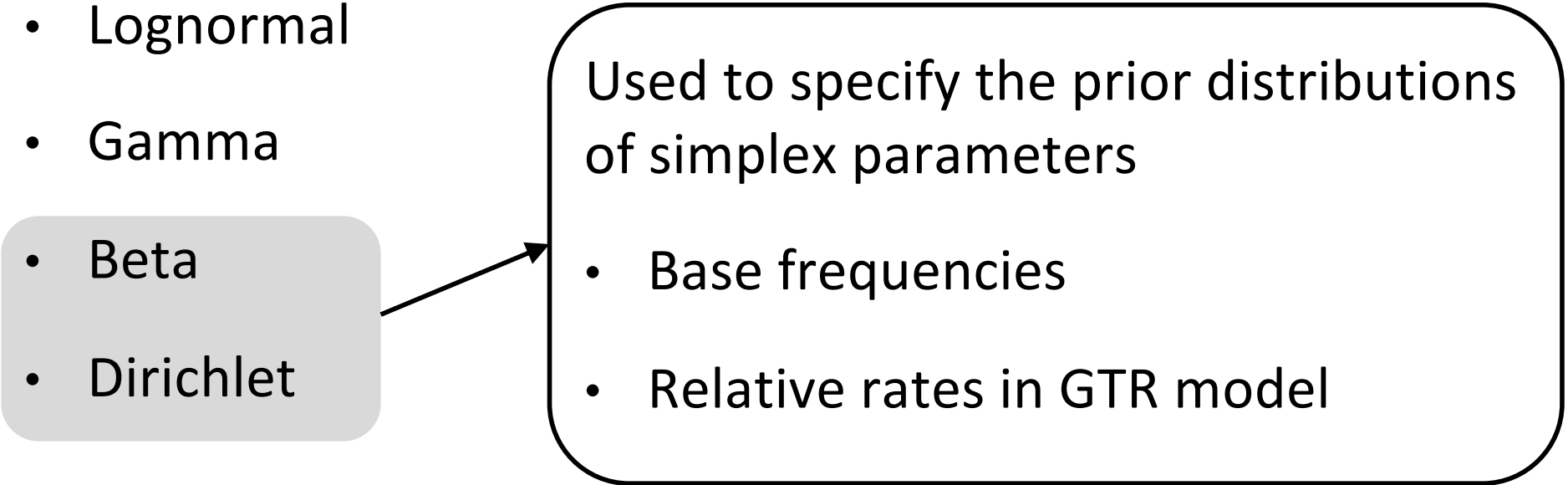
Parameters

- λ = rate of decay



Continuous distributions

- Uniform
- Normal
- Exponential
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- Dirichlet



Used to specify the prior distributions of simplex parameters

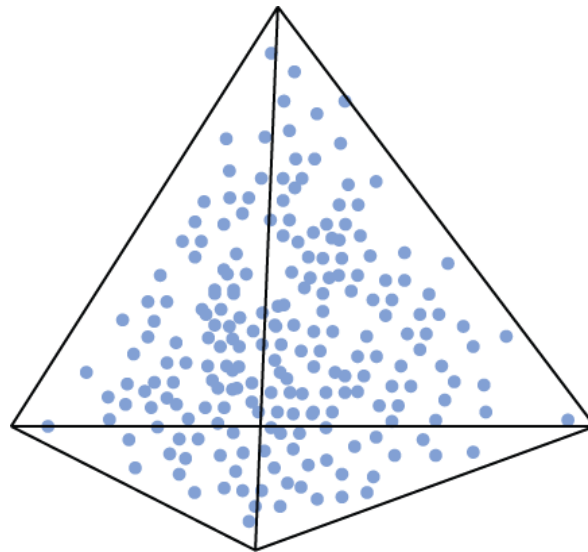
- Base frequencies
- Relative rates in GTR model

Continuous distributions

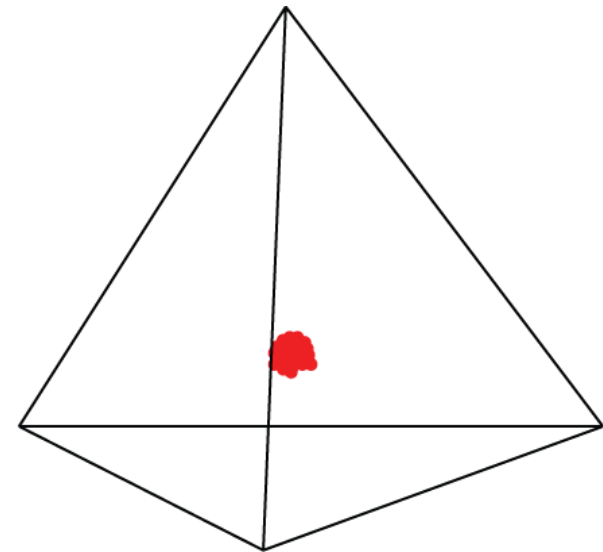
- Uniform
- Normal
- Exponential
- Lognormal
- Gamma
- Beta
- **Dirichlet**

Parameters

- $\alpha_1, \alpha_2, \dots$ = shape parameters



$$\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 1$$



$$\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 300$$

Discrete distributions

- Bernoulli distribution
- Binomial
- Multinomial
- Poisson

Default priors

	<i>BEAST2</i>	<i>MrBayes</i>
Rate matrix parameters	Gamma(0.05,10)	Dirichlet(1,1,1,1,1,1)
Base frequencies	Uniform(0,1)	Dirichlet(1,1,1,1)
Shape parameter (α)	Exponential(1)	Exponential(2)
Proportion invariable	Uniform(0,1)	Uniform(0,1)

Can specify uninformative priors where appropriate

Tree Prior

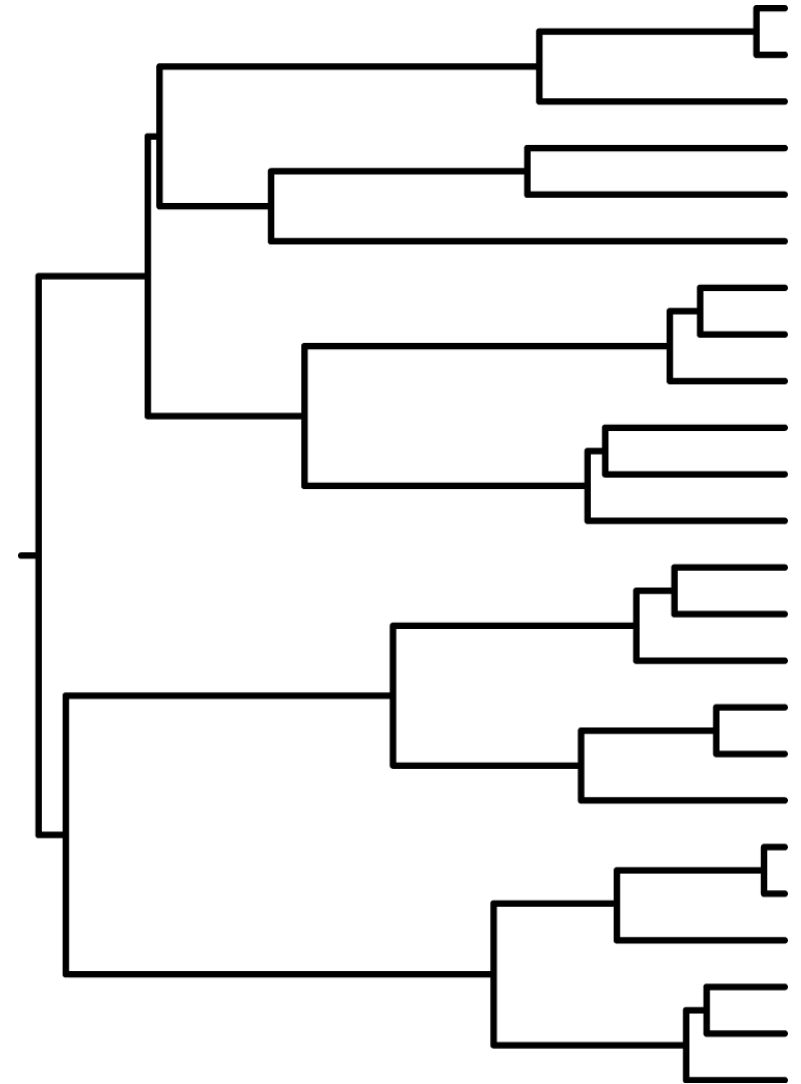
Tree prior

1. Use a **flat prior** (*MrBayes*)
 - All trees have equal probability
 - Also need a prior for branch lengths or node times

 2. Use a **biological model** (*BEAST* and *MrBayes*)
 - Among species: speciation model
 - Within species: coalescent model
- } Priors on rooted trees

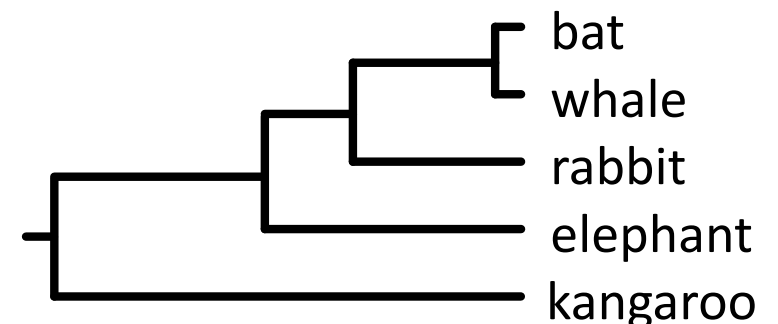
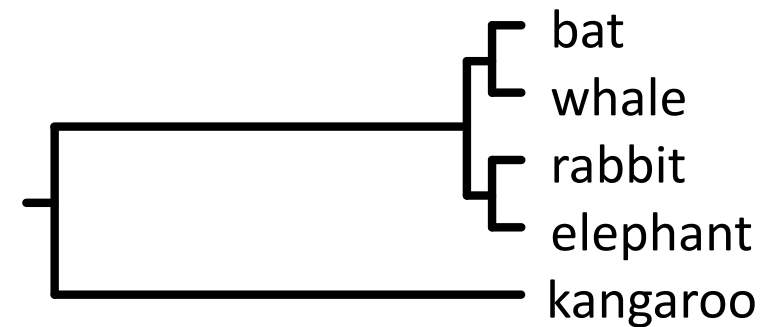
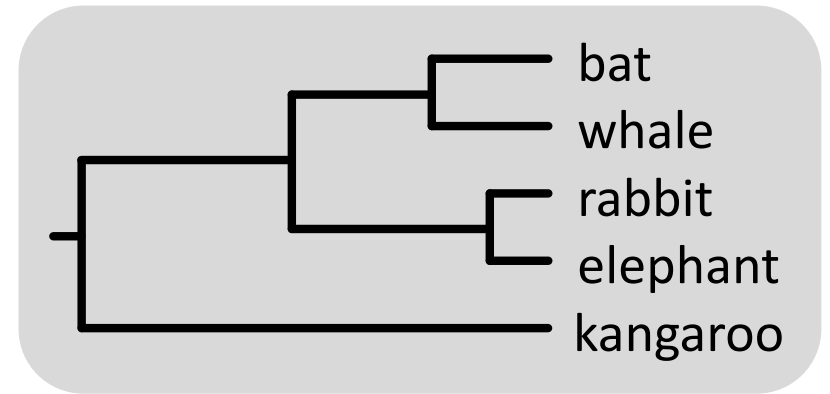
Speciation model

- Tree shape described by a stochastic branching process
- **Yule process**
 - The root lineage splits into two
 - Lineages split at a constant rate
 - Simulates speciation process
- **Birth-death process**
 - Allow lineages to go extinct



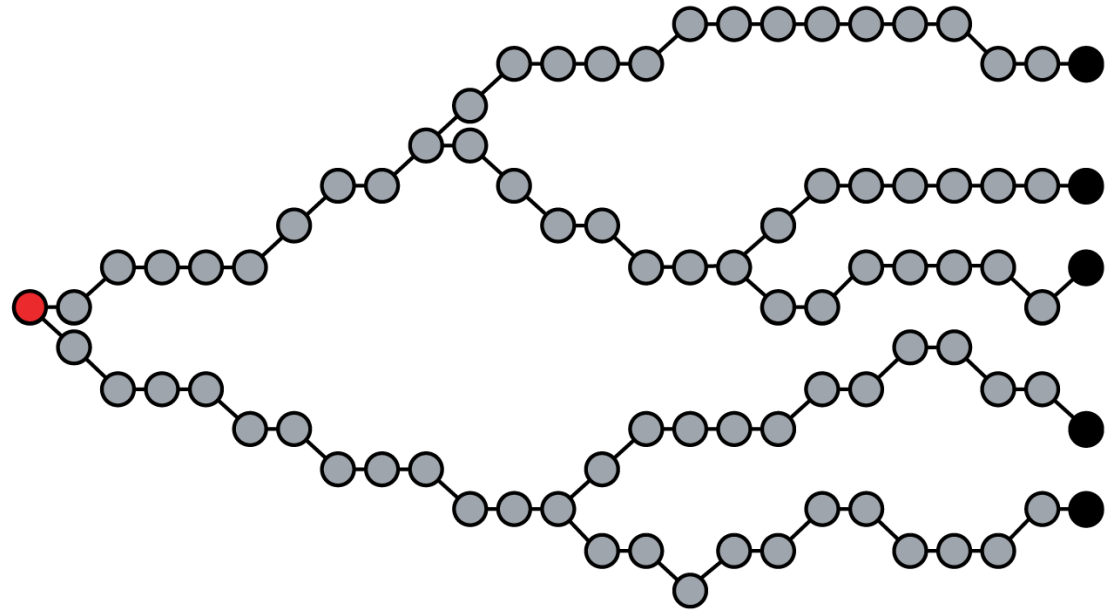
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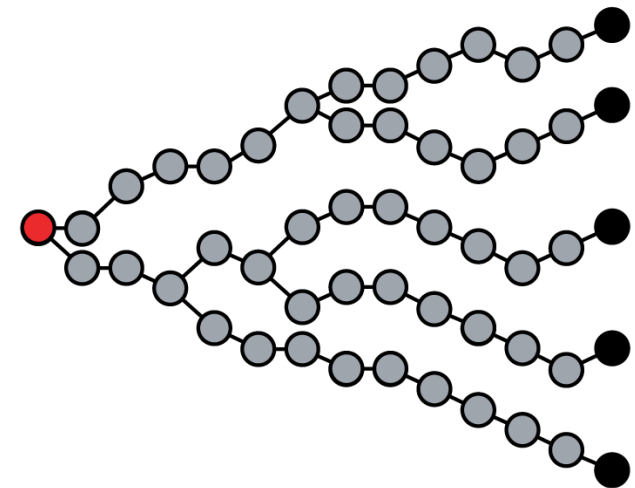


Coalescent model

Constant size



Exponential growth



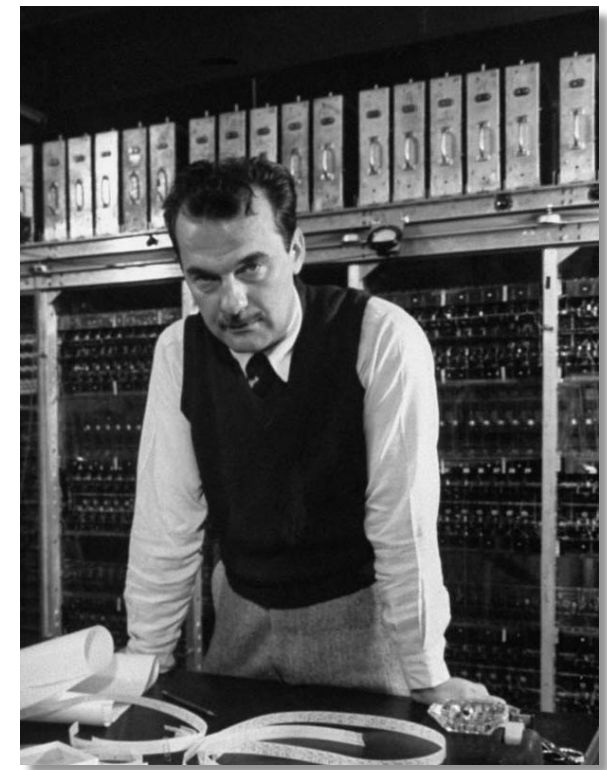
Choosing a tree prior

- Test whether inferences are robust to the choice of tree prior
- Mixed data sets: multiple sequences from each species
 - Birth-death prior generally works well
- Compare tree priors using Bayesian model selection

Posterior Distribution

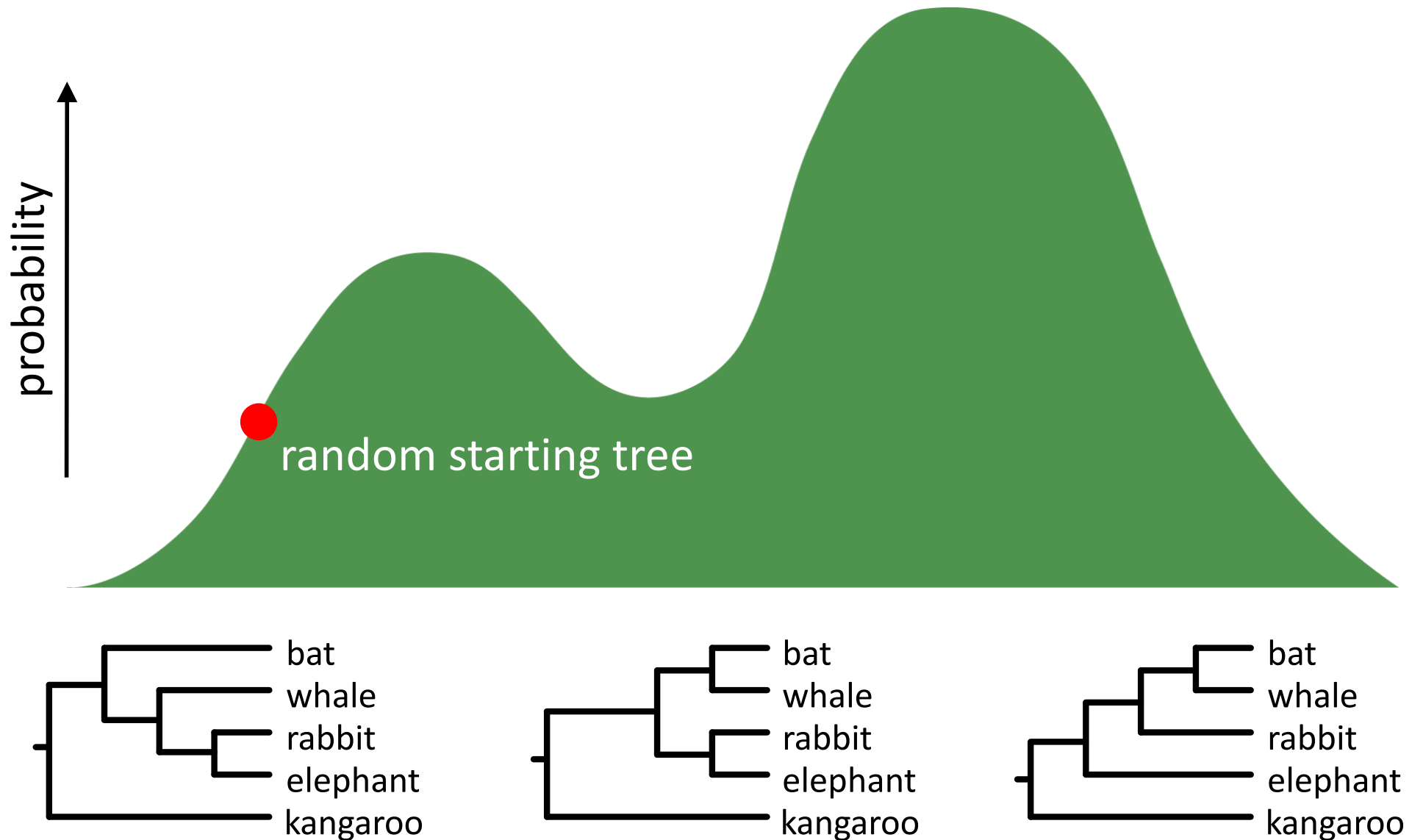
Estimating the posterior

- Impossible to obtain the posterior directly
- Instead, the posterior can be estimated using **Markov chain Monte Carlo simulation**
- This is usually done using the **Metropolis-Hastings algorithm**

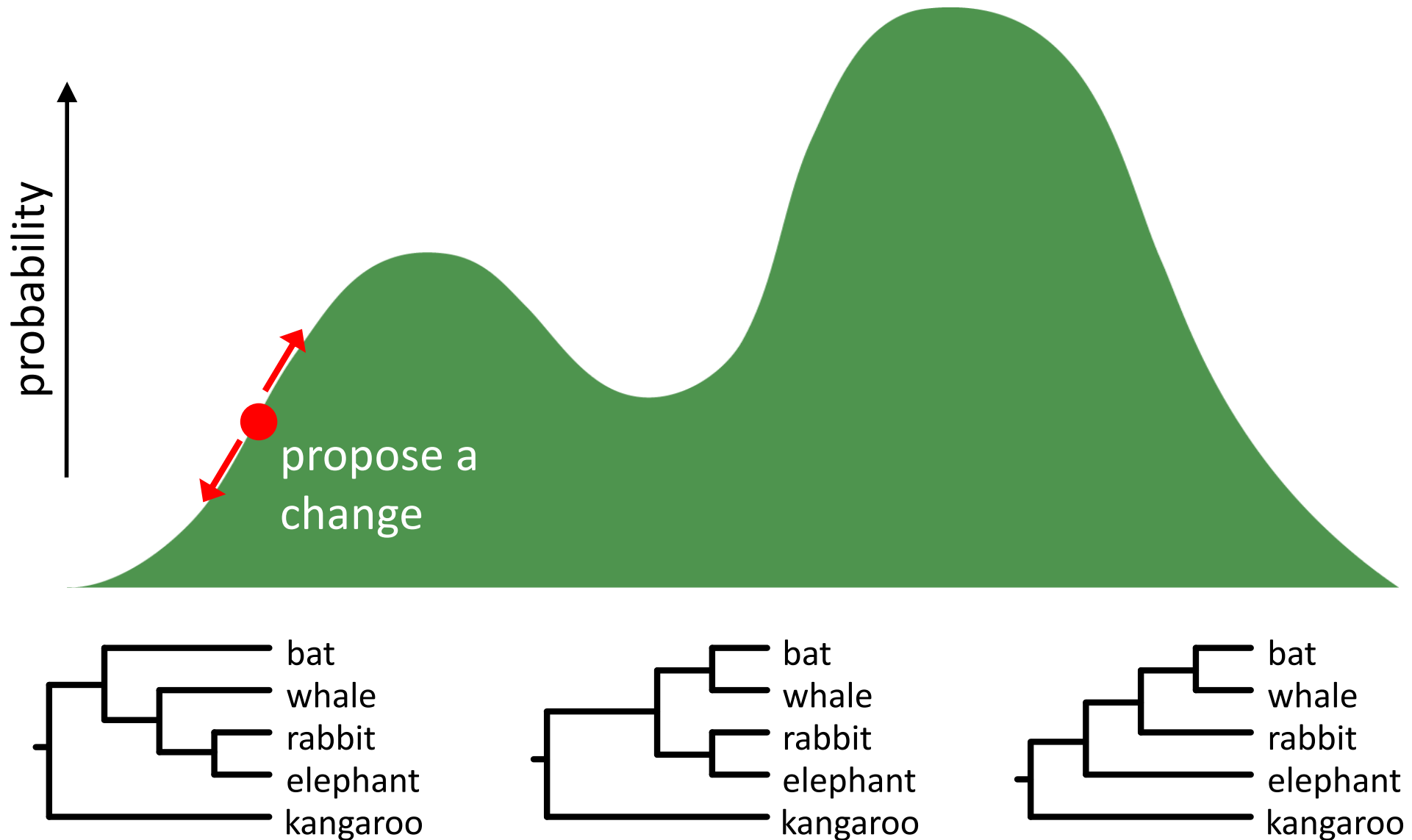


Nicholas Metropolis
Los Alamos, 1953

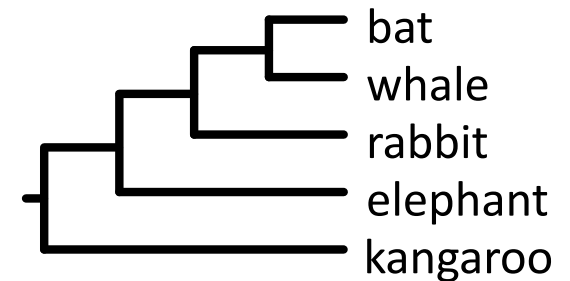
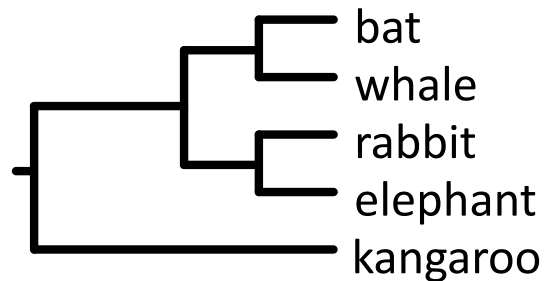
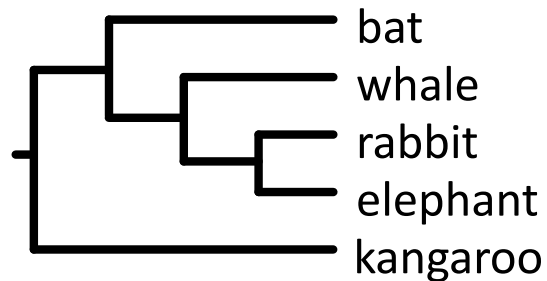
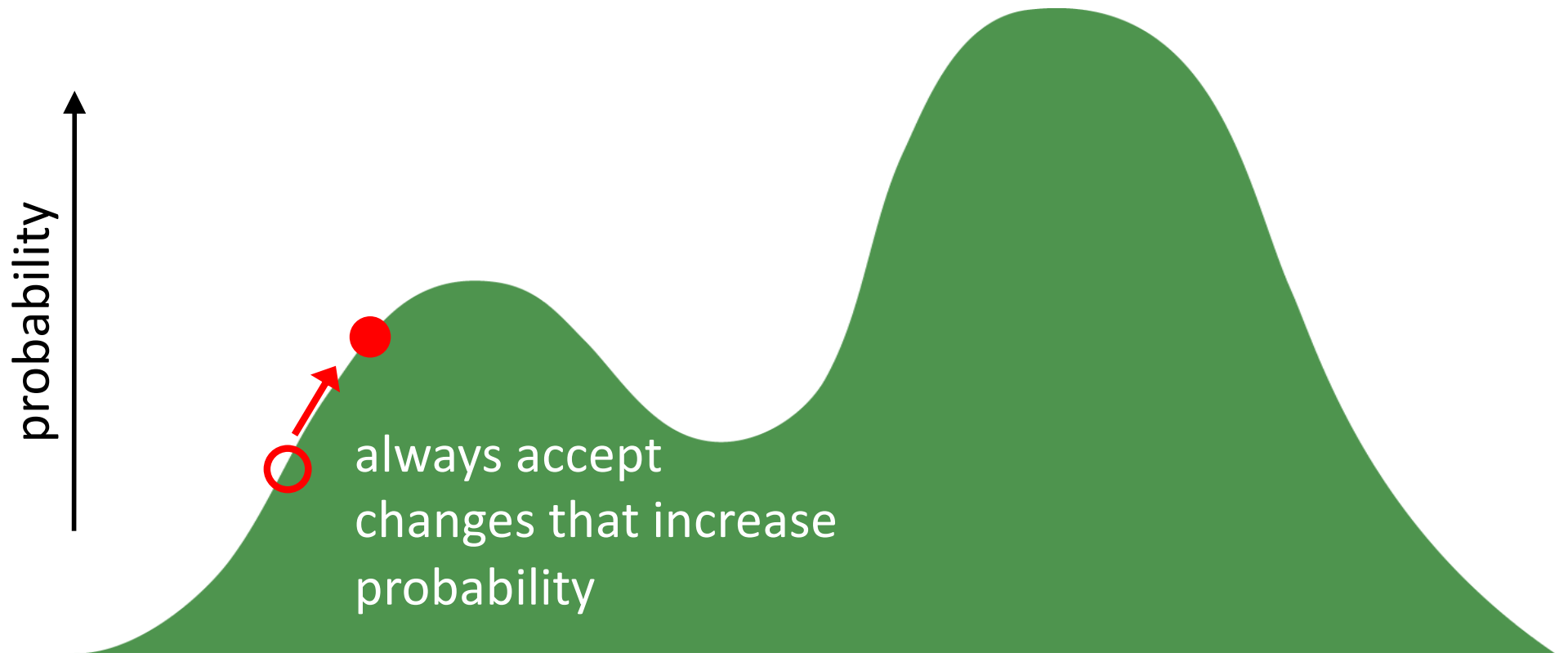
MCMC simulation



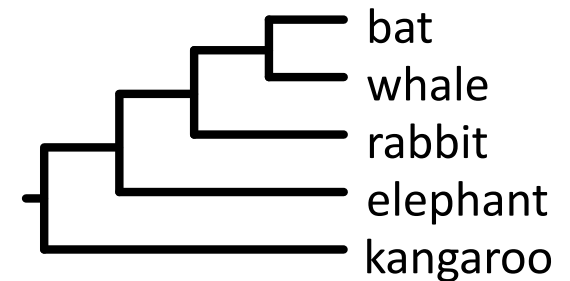
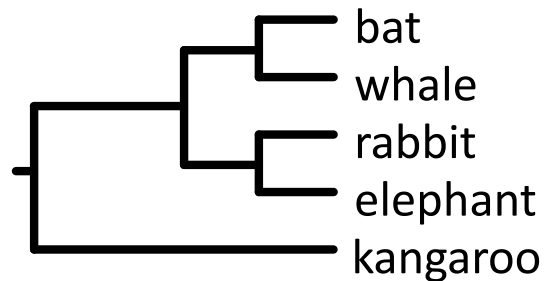
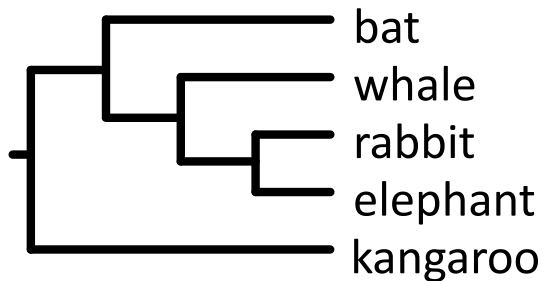
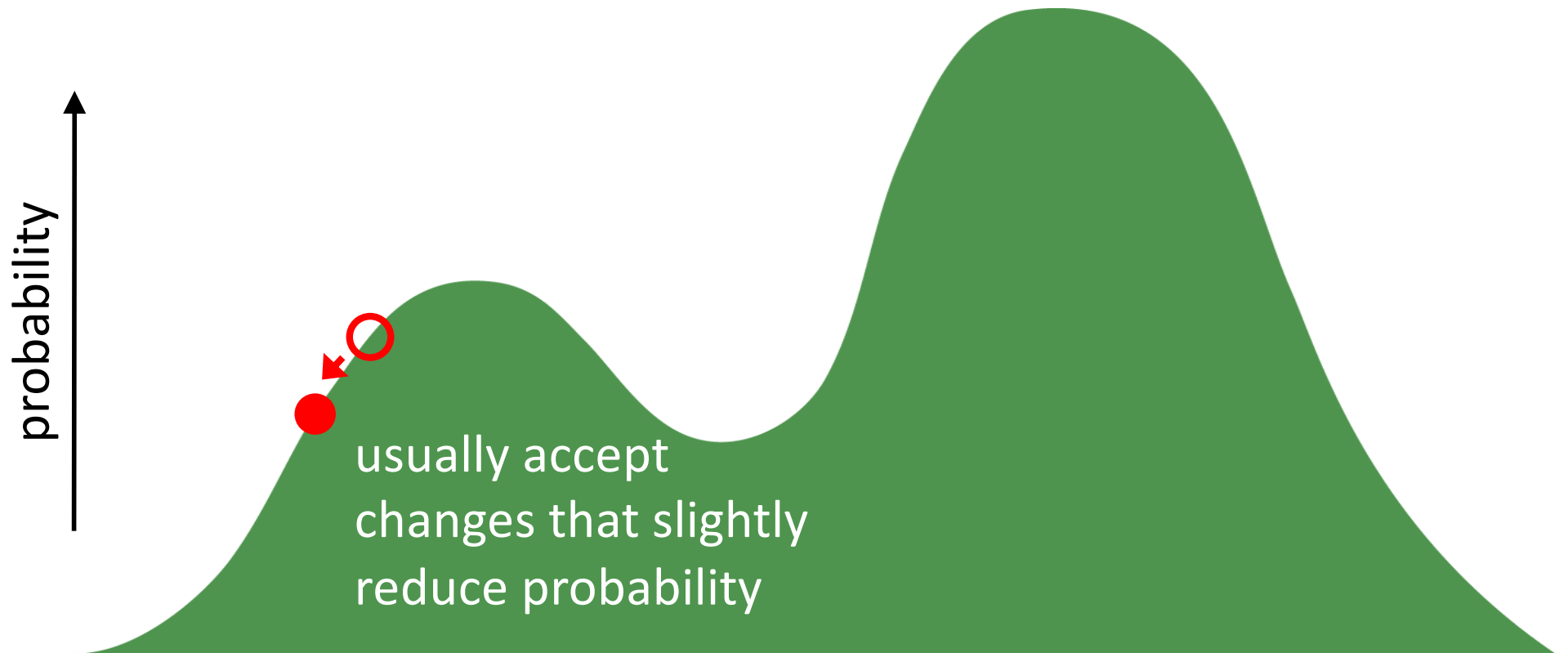
MCMC simulation



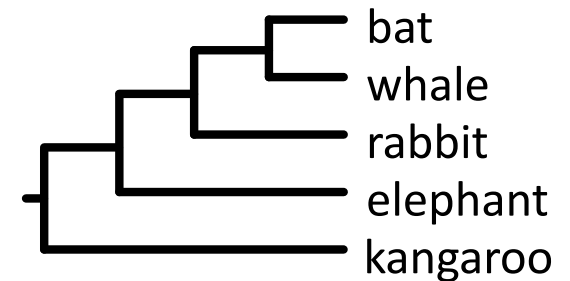
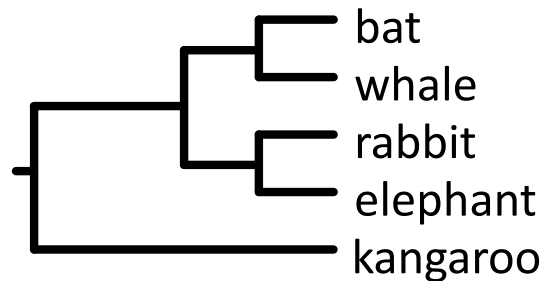
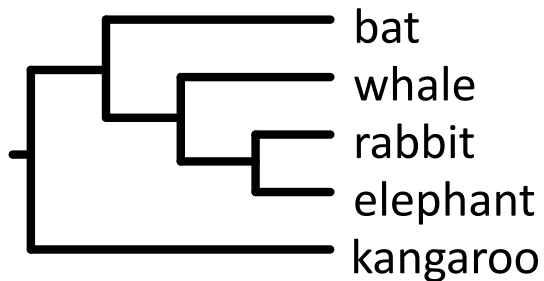
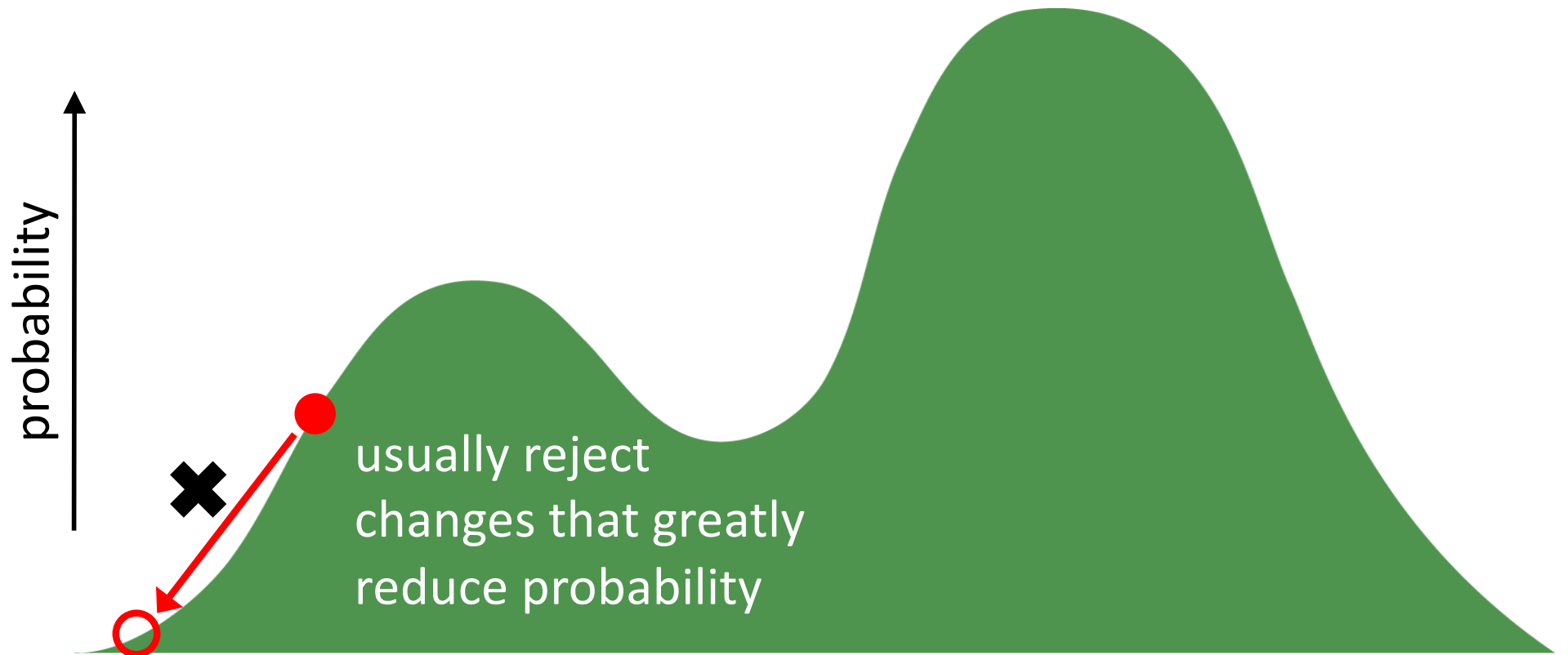
MCMC simulation



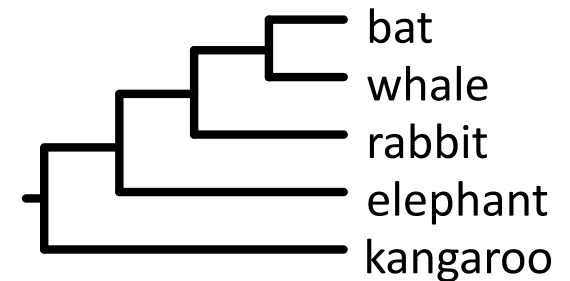
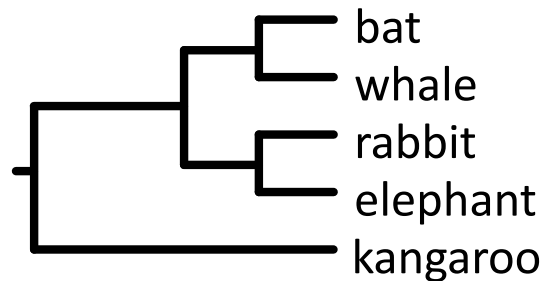
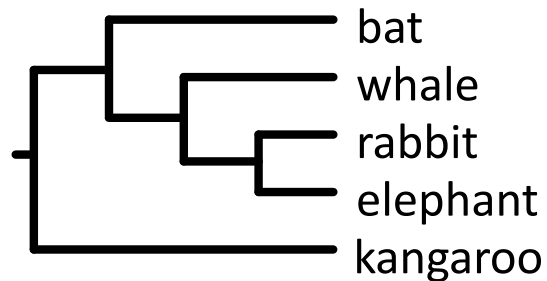
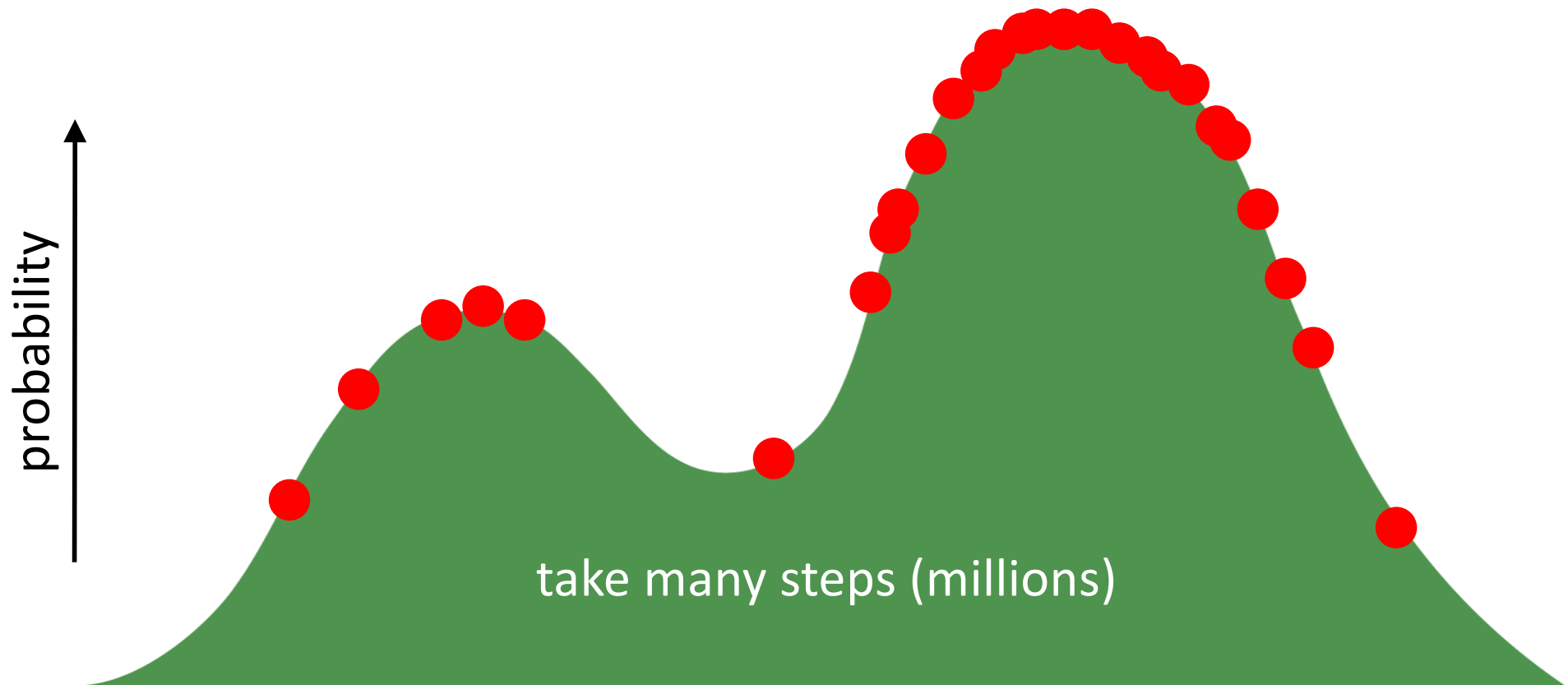
MCMC simulation



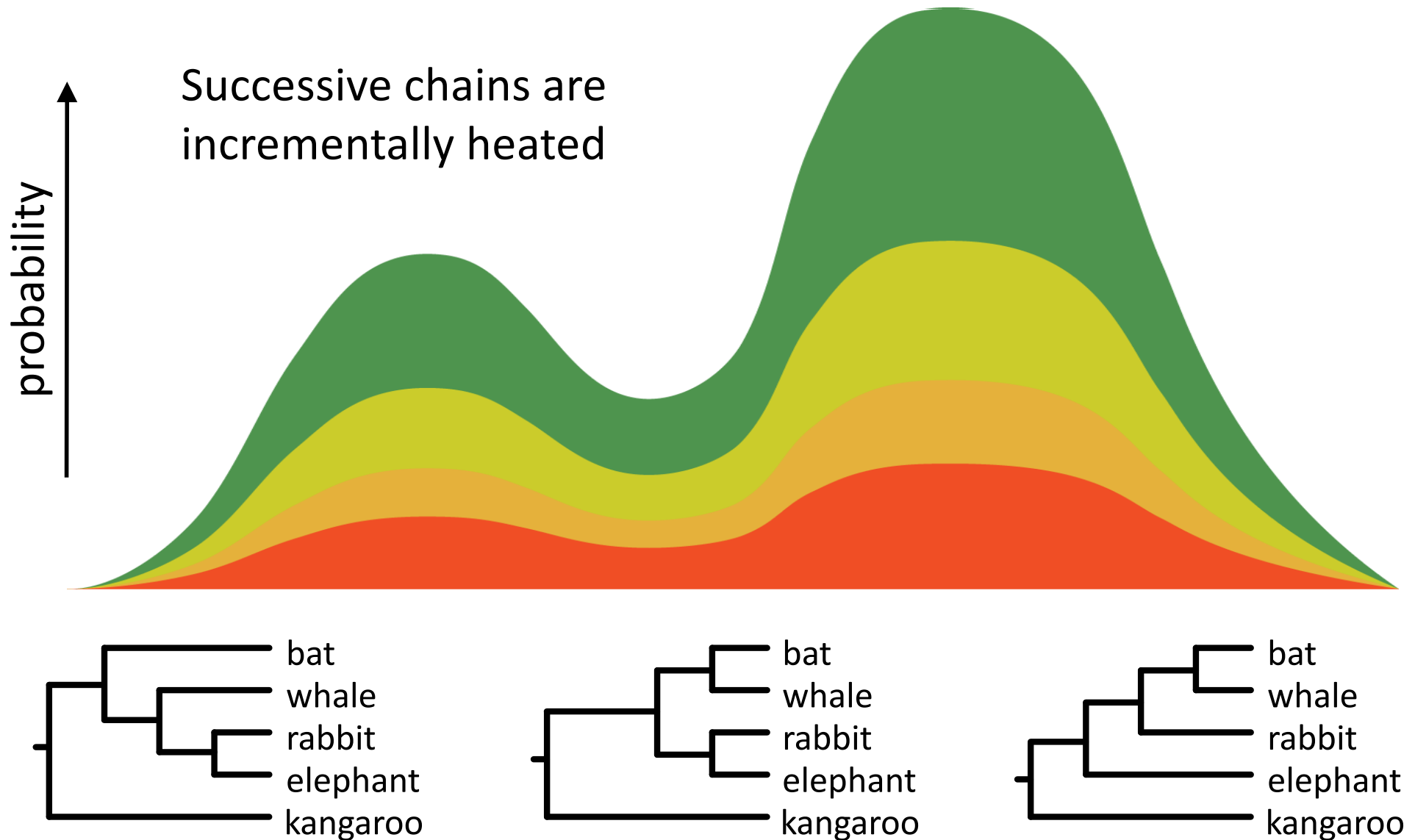
MCMC simulation



MCMC simulation



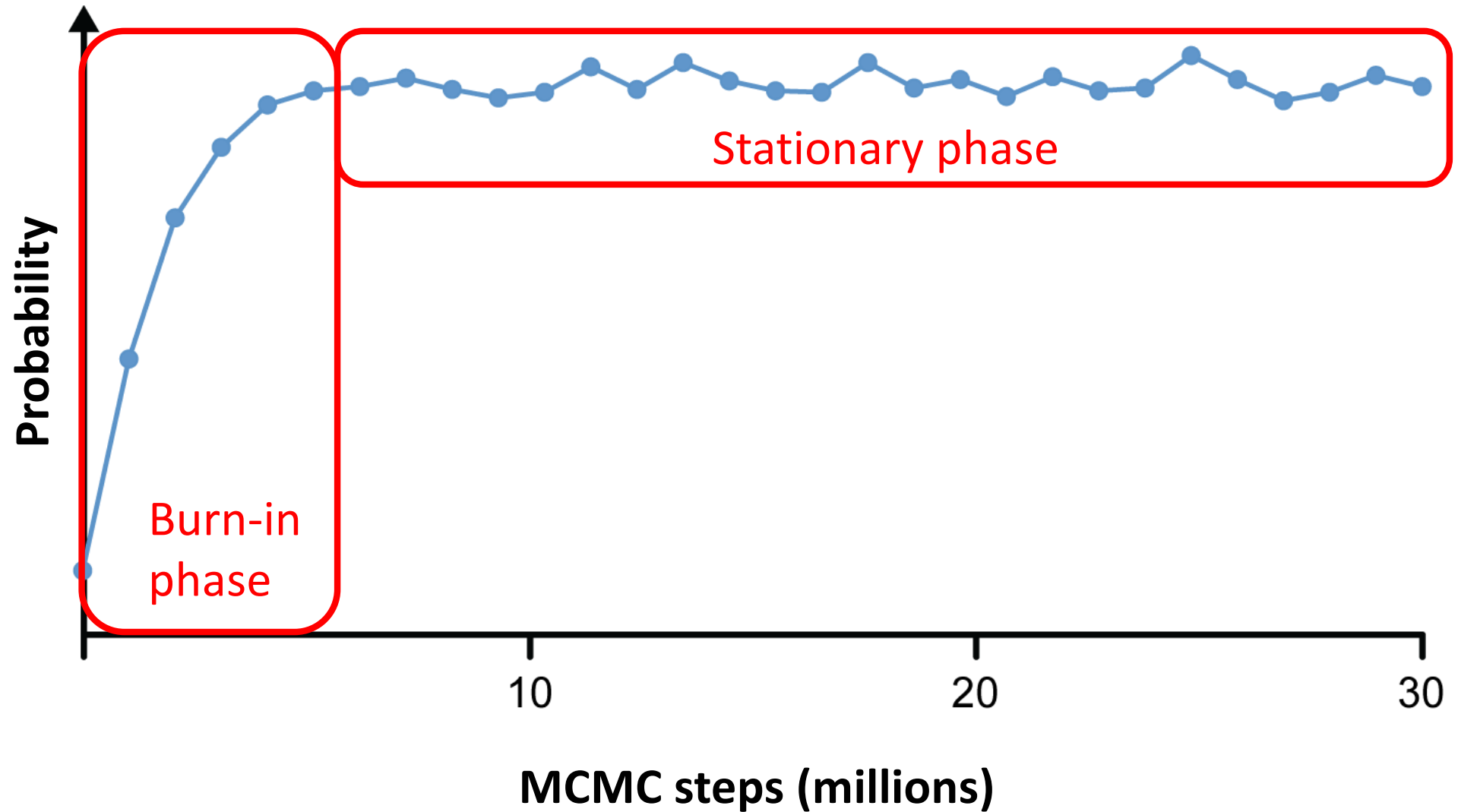
Metropolis-coupled MCMC



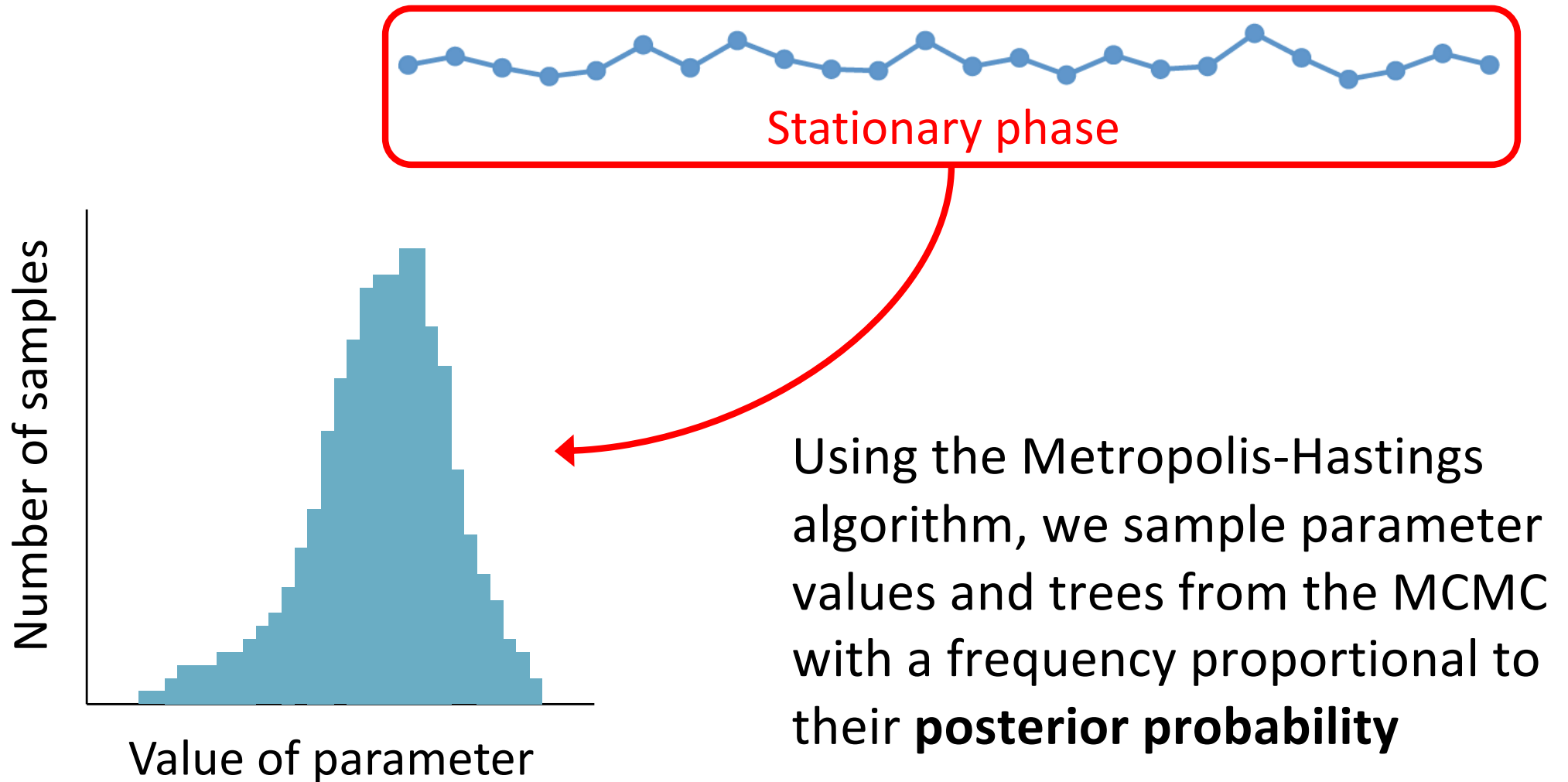
Samples from the MCMC

- Output from a Bayesian phylogenetic analysis:
 - A list of the **parameter values** visited by the Markov chain
(.p file in *MrBayes*, .log file in *BEAST*)
 - A list of the **trees** visited by the Markov chain
(.t file in *MrBayes*, .trees file in *BEAST*)

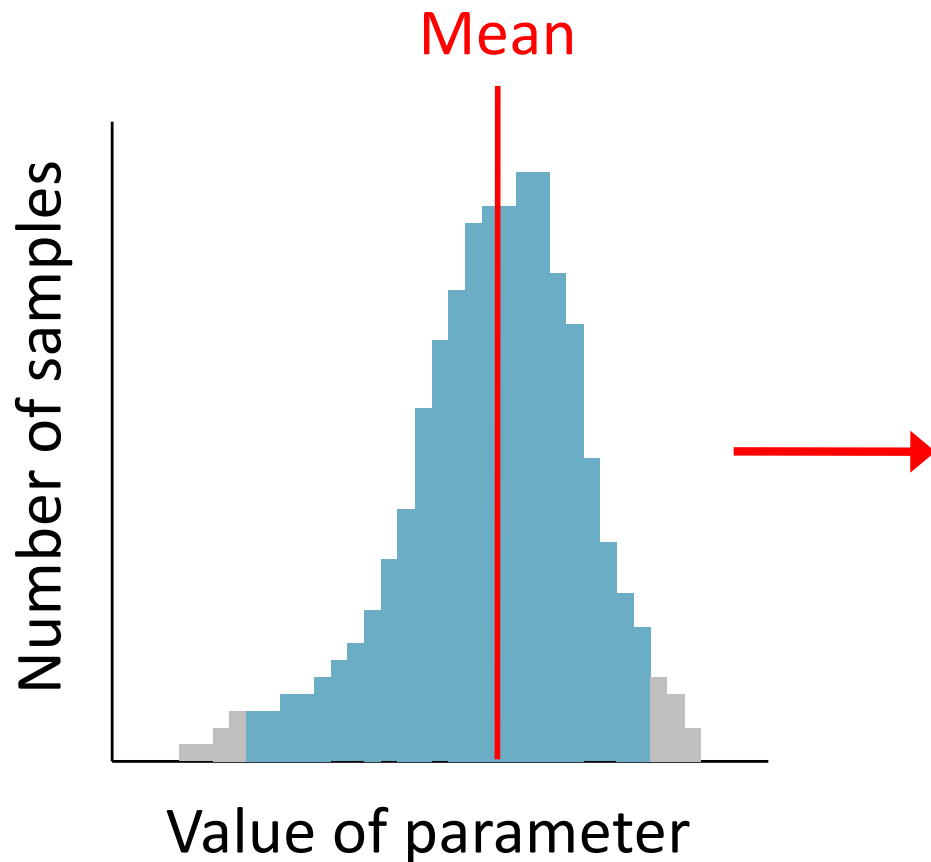
Samples from the MCMC



Samples from the MCMC



Samples from the MCMC



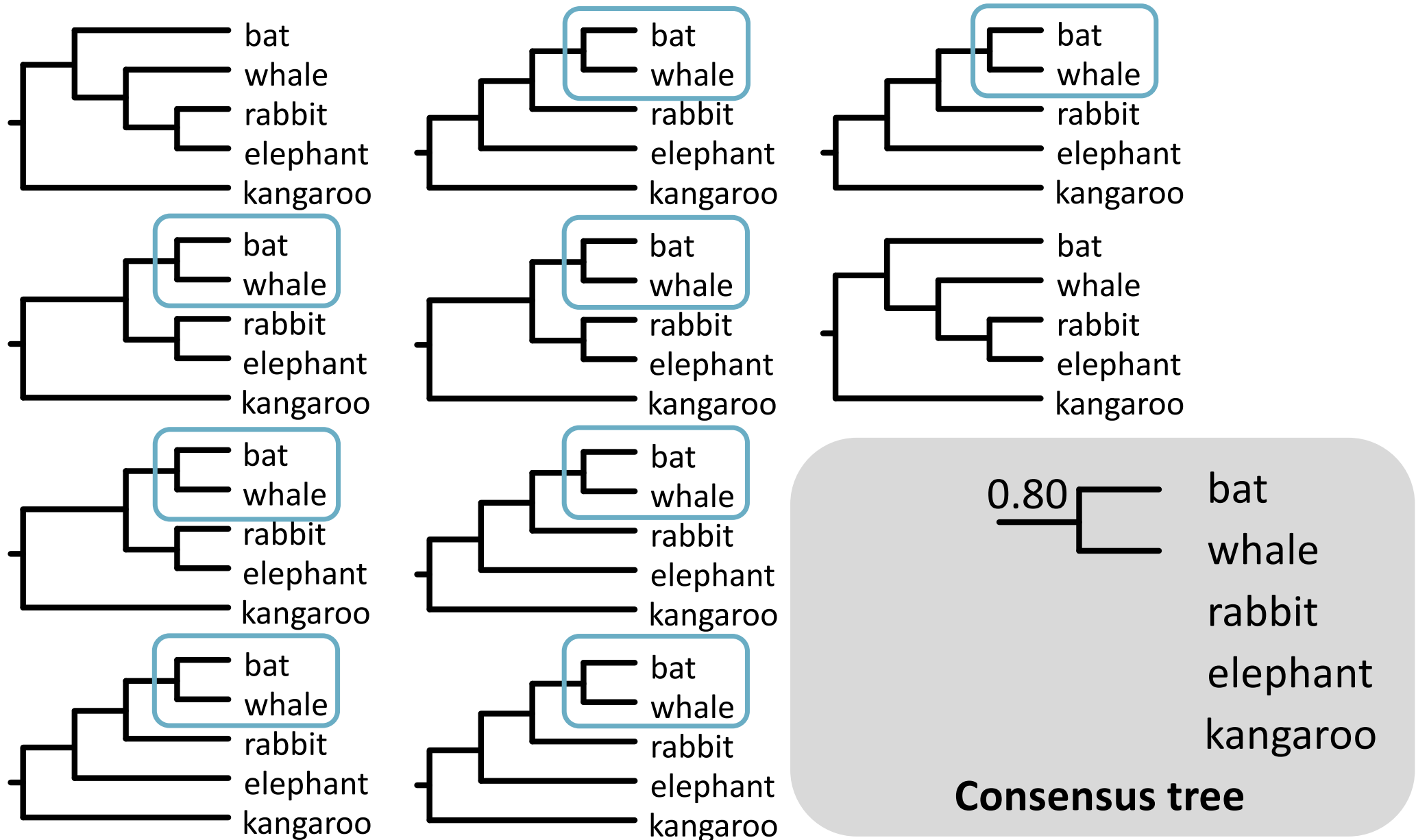
- Take the mean of the sampled values

Mean posterior estimate

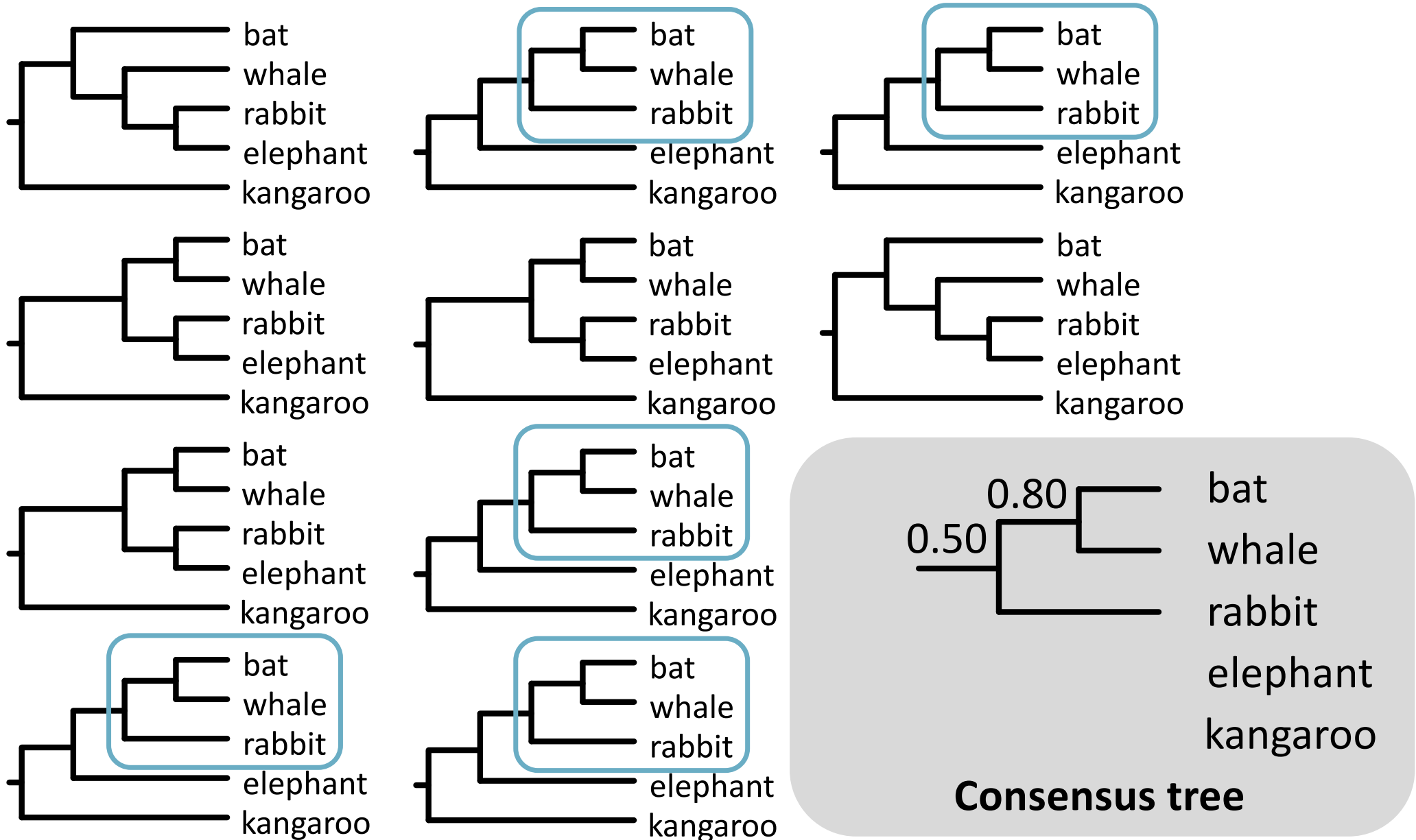
- Take the 'central' 95% of the sampled values

95% credibility interval

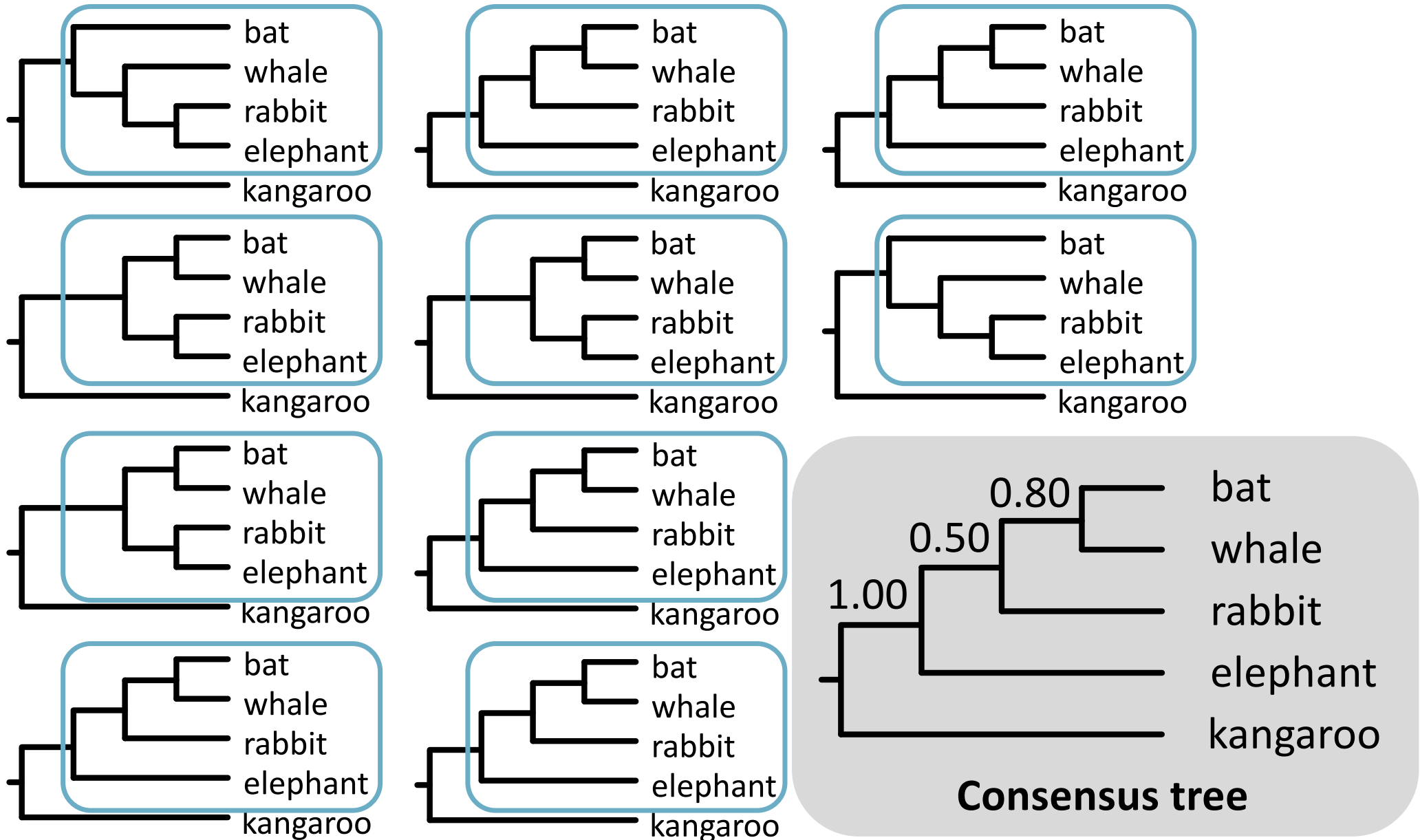
Samples from the MCMC



Samples from the MCMC



Samples from the MCMC



Samples from the MCMC

- **Majority-rule consensus tree (*MrBayes*)**
Shows all nodes with posterior probability >0.50
- **Maximum a posteriori (MAP) tree**
Sampled tree with highest posterior probability
- **Maximum clade credibility (MCC) tree (*BEAST/TreeAnnotator*)**
Sampled tree with highest sum or product of posterior node probabilities

Useful references

